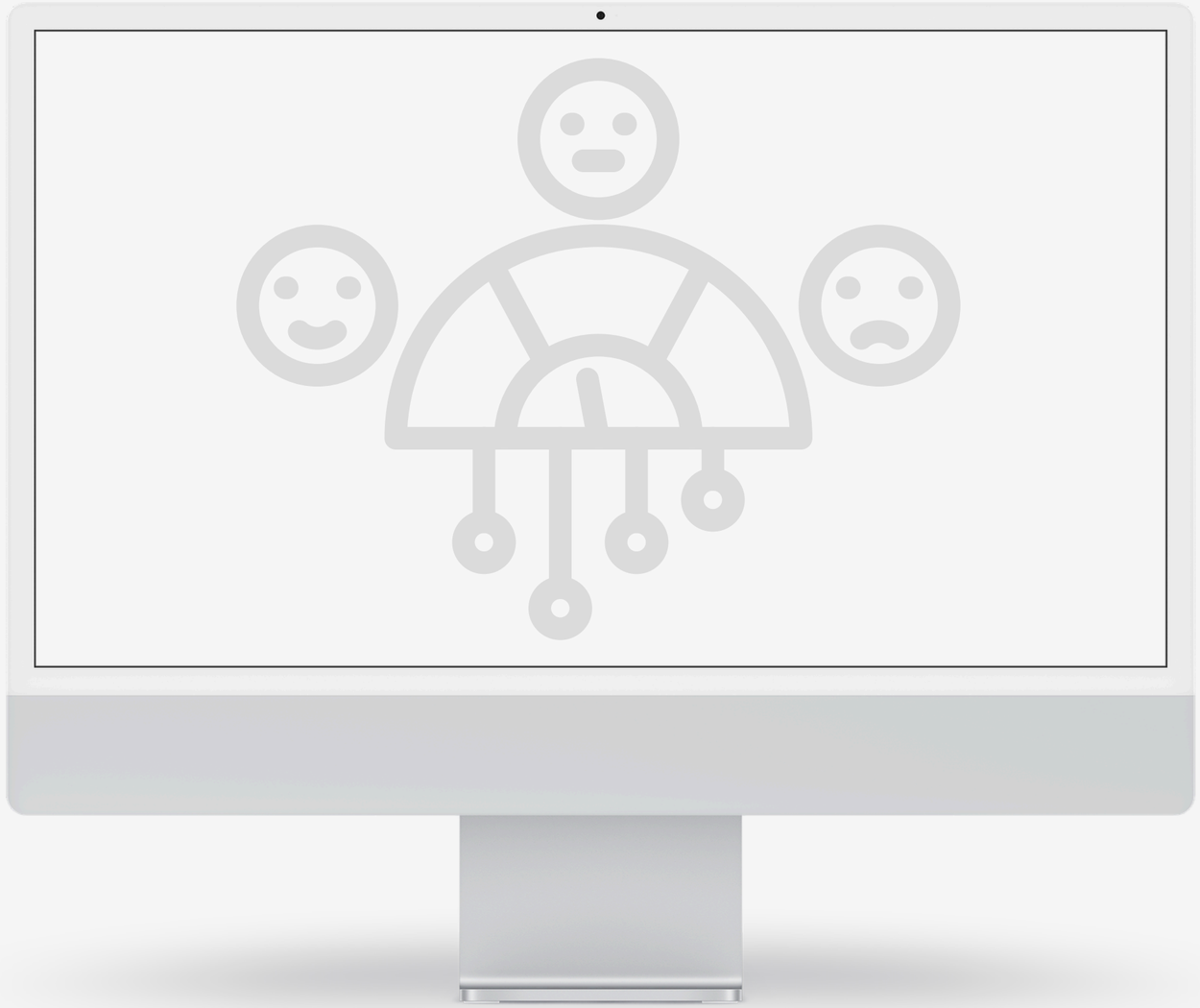
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Replication DynaSent: A Dynamic Benchmark for Sentiment Analysis



COMP8240 – Project update

Eldor Rakhmonaliev / 48756997

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THE PAPER I AM REPRODUCING

VENUE

ACL-IJCNLP 2021

THE TOP-RANKED VENUE IN COMPUTATIONAL LINGUISTICS.

AUTHORS

Potts, Wu, Geiger & Kiela

THE SAME GROUP BUILD THE STANFORD SENTIMENT TREEBANK AND ADVERSARIAL NLI.

RELEASED

Data, Code, Models

BOTH TRAINED MODELS SHIP, SO A FAITHFUL REPLICATION NEEDS NO RETRAINING.

THIS TASK IS SIMPLE TO STATE :

DECIDE WHETHER A SENTENCE IS POSITIVE, NEGATIVE OR NEUTRAL. THE INTERESTING PART IS WHAT THE PAPER SAYS ABOUT THAT THIRD LABEL.

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THE CENTRAL CLAIM

THE NEUTRAL CATEGORY IN EXISTING BENCHMARKS IS INCOHERENT

NEUTRAL THERE USUALLY MEANS A THREE-STAR RATING – NOT A JUDGEMENT A PERSON ACTUALLY MADE.

MY PROJECT TESTS WHETHER THAT CLAIM HOLDS ON DATA THE AUTHORS NEVER USED.

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HOW THE DATASET WAS BUILT

ROUND 1

HARVEST SENTENCES FROM EIGHT MILLION YELP REVIEWS, AND KEEP THE ONES A TRAINED MODEL GETS WRONG.

ROUND 2

CROWDWORKERS WRITE NEW SENTENCES ON PURPOSE, TRYING TO FOOL A STRONGER MODEL.

121, 634

SENTENCES ACROSS BOTH ROUNDS

5

WORKERS PER SENTENCE; 3 WHO AGREE SET THE GOLD LABEL

33.3

CHANCE BY CONSTRUCTION ON DEV AND TEST

THE FULL FIVE-WAY RESPONSE DISTRIBUTION SHIPS WITH THE DATA, NOT JUST THE MAJORITY – AND THAT TURNS OUT TO MATTER.

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HOW MY REPLICATION RUNS

BOTH MODELS ARE RoBERTa – BASE WITH A THREE-WAY CLASSIFIER HEAD. MODEL 0 IS TRAINED WITHOUT DYNASENT; MODEL 1 IS TRAINED WITH IT.

I WROTE ONE SCRIPT PER TABLE. EACH PRINTS MY OUTPUT BESIDE THE PUBLISHED FIGURE AND THE DIFFERENCE, SO NO NUMBER IS RETYPED BY HAND.

EVERY TABLE IS WRITTEN AS CSV AND LaTeX, SO THE REPORT READS ITS FIGURES STRAIGHT OUT OF THE PIPELINE.

```
PS> .\replication\run_all.ps1
[1/7] check_setup          _ _ _ _ _ ok
[2/7] dataset_tables       _ _ _ _ _ ok
[3/7] human_performance    _ _ _ ok
[4/7] example_tables       _ _ _ _ ok
[5/7] external_data        _ _ _ _ _ ok
[6/7] evaluate             _ _ _ _ _ _ ok
[7/7] new_data_eval        _ _ _ _ ok
```

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WHERE THE REPLICATION LANDED

TABLE	WHAT IT REPORTS	RESULT
4, 7, 10	DATASET COMPOSITION, BOTH ROUNDS	EXACT
6, 9	HUMAN PERFORMANCE AND FLEISS' KAPPA	EXACT, TO 3 DECIMALS
3, 8	WORKED EXAMPLES FROM EACH ROUND	EXACT
1	EXTERNAL TRAINING CORPORA, 24 CELLS	24 / 24
2B, 5B	MODEL SCORES ON EVERY EVALUATION SET	EXACT EXCEPT AMAZON

TWO THINGS DID NOT MATCH AT FIRST. BOTH WERE WORTH CHASING.

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A SINGLE SPACE WAS COSTING 33 POINTS

WHAT MY SENTENCES SAID

THE FOOD WAS GOOD .

WHAT THE MODEL WAS TRAINED ON

THE FOOD WAS GOOD.

12.6

SST-3 DEV NEUTRAL F1 - BEFORE

45.7

AFTER THE FIX

45.4

PUBLISHED IN THE PAPER

ROBERTA'S TOKENISER IS WHITESPACE-SENSITIVE, SO THOSE ARE DIFFERENT INPUTS. A PLAN JOIN MATCHED 14 OF 1, 101 SENTENCES; MY DETOKENISER MATCHED 1, 011.

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AMAZON STILL RUNS ONE TO TWO POINTS HIGH

MY SCORES MINUS THE PUBLISHED ONES

+0. 8 TO + 2. 2

CONSISTENT, AND ALL IN ONE DIRECTION – A CONSTRUCTION DIFFERENCE, NOT NOISE.

THE HYPOTHESIS

ZHANG'S AMAZON FILES KEEP TITLE AND CONTENT IN SEPERATE COLUMNS. I JOIN THEM AS "TITLE CONTENT". THE PAPER DOES NOT SAY WHICH IT USED.

THE TEST

REBUILD THE EVALUATION SET FROM CONTENT ONLY AND RE-SCORE. ONE RUN SETTLES IT.

DOCUMENTED AS AN OPEN ITEM RATHER THAN QUIETLY ROUNDED AWAY.

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THE NEUTRAL GAP CLOSES FROM 19.6 TO 1.4

TWELWE THOUSAND TWEETS WHERE ANNOTATORS CHOSE NEUTRAL THEMSELVES, RATHER THAN IT BEING DERIVED FROM A STAR RATING. THE NEUTRAL GAP IS POSITIVE AND NEGATIVE PERFORMANCE MINUS NEUTRAL PERFORMANCE.

MODEL	POS & NEG, MEAN F1	NEUTRAL F1	NEUTRAL GAP
MODEL 0 - TRAINED WITHOUT DYNASENT	59.2	39.6	19.6
MODEL 1 - TRAINED WITH DYNASENT	66.1	64.7	1.4
CHANGE (POSITIVE +8.6, NEGATIVE +5.4)	7	25.1	-18.2

ON THE STAR RATED CORPORA, MODEL 1'S GAP STAYS BETWEEN 23 AND 44. THE BENEFIT TRANSFERS WHERE A HUMAN CHOSE THE WORD NEUTRAL – NOT WHERE NEUTRAL MEANS THREE STARS.

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GOOGLE LOCAL, FROM UC SAN DIEGO

666M

GOOGLE MAPS REVIEWS, PUBLICLY RELEASED FOR RESEARCH

2021

COVERED THROUGH SEPTEMBER – LATER THAN ANYTHING THESE MODELS SAW

0

API TERMS TO BREACH: A RESEARCH RELEASE, CITED, NOT SCRAPED

500, 000 REVIEWS > 720, 892 SENTENCES > FILTER & DUPLICATES > 2, 000 BALANCED PROMPTS

A DIFFERENT PLATFORM, A DIFFERENT REGION AND A LATER PERIOD THAN YELP. IF THE PAPER'S ARGUMENT IS ABOUT SENTIMENT RATHER THAN ABOUT YELP, IT SHOULD SURVIVE THE MOVE

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THREE-STAR IS CONTRADICTED THREE TIMES OUT OF FOUR

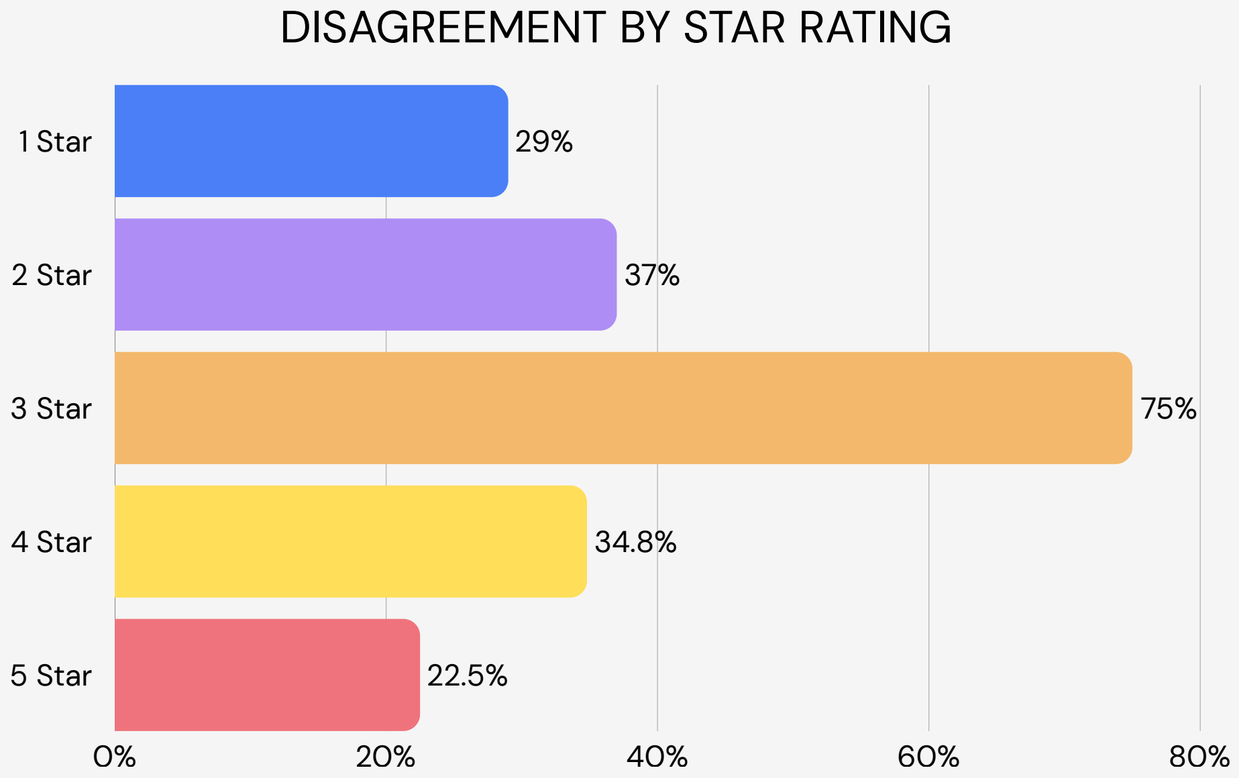
OVERAL DISAGREEMENT

39.8%

796 OF 2000 SENTENCES

THE PAPER'S OWN ROUNDS

ROUND 1	46.5%
ROUND 2	18.7%



THE ENDS OF THE SCALE ARE EASY. THE MIDDLE IS NOT A SENTIMENT CATEGORY – THE PAPER'S ARGUMENT, REPRODUCED INDEPENDENTLY ON A PLATFORM THEY NEVER TOUCHED.

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FROM CANDIDATES TO A VALIDATED ROUND THREE

NEXT

VALIDATION

SEVERAL INDEPENDENT JUDGEMENTS PER SENTENCE; THE MAJORITY BECOMES THE GOLD LABEL – THE PAPER'S SECTION 3.3 , WITH ITS ORIGINAL INSTRUCTION

THEN

CALIBRATION

RUN THE SAME JUDGES OVER DYNASENT'S OWN DEVELOPEMENT SETS, WHERE THE HUMAN ANSWERS ARE ALREADY KNOWN.

AND

CLOSE THE AMAZON GAP

REBUILD THE EVALUATION SET FROM CONTENT ONLY, RE-SCORE, AND WRITE IT UP EITHER WAY.

EVERTHING IS IN THE REPOSITORY

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THANK YOU